

Bharat L Bhuva

其他

基本信息

姓名: Bharat L Bhuva

所属院校: 范德堡大学

地区: United States

学科领域

微电子科学与工程

集成电路设计与集成系统

研究领域

微电子器件与制造

数字集成电路设计

纳米电子与先进工艺

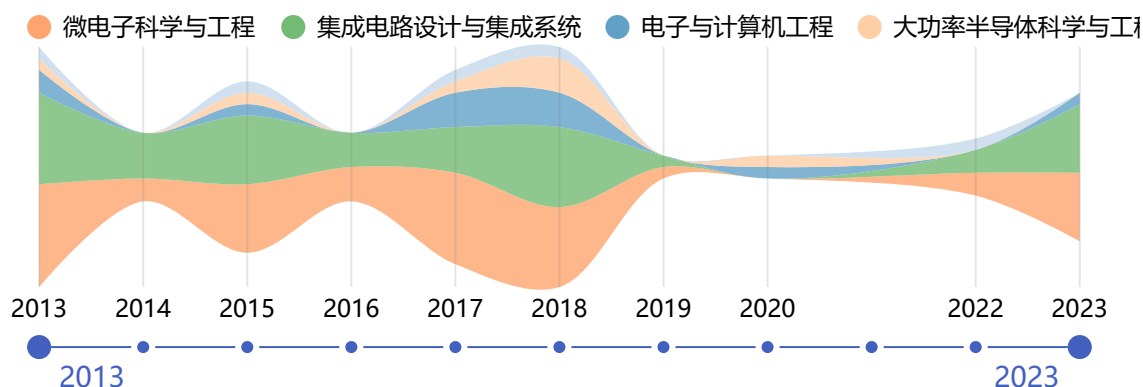
EDA工具与电路验证

低功耗电路与能量管理

个人简介

Dr. Bharat L Bhuva is a distinguished expert in the field of Computer-Aided Design (CAD) tools, with a profound focus on radiation effects on integrated circuits, modeling of semiconductor devices, and VLSI design. His extensive research also encompasses the intricate processes involved in semiconductor fabrication. Dr. Bhuva's contributions significantly advance the understanding and application of integrated circuit technology and semiconductor engineering.

近年发表的论文



Efficacy of Transistor Stacking on Flip-Flop SEU Performance at 22-nm FDSOI Node

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Li, Zongru; Elash, Christopher; Jin, Chen; Chen, Li; Wen, Shi-Jie; Fung, Rita; Xing, Jiesi; Shi, Shuting; Yang, Zhi Wu; Bhuva, Bharat L.

被引用次数: 4

2023

Single-Event Upsets for Single-Port and Two-Port SRAM Cells at the 5-nm FinFET Technology

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Pieper, N. J.; Xiong, Y.; Pasternak, J.; Dodds, N. A.; Ball, D. R.; Bhuva, B. L.

被引用次数: 1

2023

Evaluation of the Single-Event-Upset Vulnerability for Low-Energy Protons at the 7-and 5-nm Bulk FinFET Nodes

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Xiong, Yoni; Pieper, Nicholas J.; McCurdy, Michael W.; Ball, Dennis R.; Sierwaski, Brian D.; Bhuva, Bharat L.

被引用次数: 2

2023

SEU Performance of RHBD Flip-Flops Using Guard Gates at 22-nm FDSOI Technology Node

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Li, Zongru; Elash, Christopher; Xing, Jiesi; Jin, Chen; Chen, Li; Wen, Shi-Jie; Fung, Rita; Shi, Shuting; Yang, Zhi Wu; Bhuva, Bharat L.

被引用次数: 3

2023

Impact of CnRx Structure on Soft Error Rates of Flip-Flop Designs at 22-nm FD SOI Node

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Elash, Christopher J.; Li, Zongru; Jin, Chen; Chen, Li; Wen, Shi-Jie; Fung, Rita; Xing, Jiesi; Shi, Shuting; Yang, Zhi Wu; Bhuva, Bharat L.

被引用次数: 1

2023

Efficacy of Spatial and Temporal RHBD Techniques at Advanced Bulk FinFET Technology Nodes

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Xiong, Yoni; Pieper, Nicholas J.; Narasimham, Balaji; Ball, Dennis R.; Bhuva, Bharat L.

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2023

Single-Event Upset Cross-Section Trends for D-FFs at the 5-and 7-nm Bulk FinFET Technology Nodes

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Xiong, Yoni; Pieper, Nicholas J.; Feeley, Alexandra T.; Narasimham, Balaji; Ball, Dennis R.; Bhuva, Bharat L.

被引用次数: 4

2023

Total-Ionizing-Dose Effects and Low-Frequency Noise in N-Type Carbon Nanotube Field-Effect Transistors With HfO₂ Gate Dielectrics

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Darmawi-Iskandar, Patrick K.; Aaron, Andrew M.; Zhang, En Xia; Bhuva, Bharat L.; Kauppila, Jeffery S.; Davidson, Jim L.; Alles, Michael L.; Fleetwood, Daniel M.; Massengill, Lloyd W.

被引用次数: 2

2023

Modeling Logic Error Single-Event Cross Sections at the 7-nm Bulk FinFET Technology Node

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Xiong, Yoni; Feeley, Alexandra T.; Ball, Dennis R.; Bhuva, Bharat L.

被引用次数: 2

2022

Effect of Frequency on Total Ionizing Dose Response of Ring Oscillator Circuits at the 7-nm Bulk FinFET Node

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Feeley, Alex; Xiong, Yoni; Guruswamy, Nithin; Bhuva, B. L.

被引用次数: 10

2022

SE Performance of D-FF Designs With Different VT Options at Near-Threshold Supply Voltages in 7-nm Bulk FinFET Technology

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Feeley, A.; Xiong, Y.; Pieper, N. J.; Ball, D. R.; Bhuva, B. L.

被引用次数: 2

2022

SE Response of Guard-Gate FF in 16-and 7-nm Bulk FinFET Technologies

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Cao, Jingchen; Xu, Lyuan; Bhuva, Bharat L.; Fung, Rita; Wen, Shi-Jie; Cazzaniga, Carlo; Frost, C.

被引用次数: 6

2020

Exploiting SEU Data Analysis to Extract Fast SET Pulses

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Harrington, R. C.; Kauppila, J. S.; Maharrey, J. A.; Haeffner, T. D.; Zhang, E. X.; Ball, D. R.; Nsengiyumva, P.; Alles, M. L.; Bhuva, B. L.; Massengill, L. W.

被引用次数: 1

2019

Empirical Modeling of FinFET SEU Cross Sections Across Supply Voltage

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Harrington, R. C.; Kauppila, J. S.; Maharrey, J. A.; Haeffner, T. D.; Sternberg, A. L.; Zhang, E. X.; Ball, D. R.; Nsengiyumva, P.; Bhuva, B. L.; Massengill, L. W.

被引用次数: 10

2019

Power-Aware SE Analysis of Different FF Designs at the 14-/16-nm Bulk FinFET CMOS Technology Node

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Jiang, H.; Zhang, H.; Chatterjee, I.; Kauppila, J. S.; Bhuva, B. L.; Massengill, L. W.

被引用次数: 6

2018

Analysis of Temporal Masking Effects on Master- and Slave-Type Flip-Flop SEUs and Related Applications

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Chen, R. M.; Mahatme, N. N.; Diggins, Z. J.; Wang, L.; Zhang, E. X.; Chen, Y. P.; Liu, Y. N.; Narasimham, B.; Witulski, A. F.; Bhuva, B. L.; Fleetwood, D. M.

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2018

Dual-Interlocked Logic for Single-Event Transient Mitigation

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Maharrey, J. A.; Kauppila, J. S.; Harrington, R. C.; Nsengiyumva, P.; Ball, D. R.; Haeffner, T. D.; Zhang, E. X.; Bhuva, B. L.; Holman, W. T.; Massengill, L. W.

被引用次数: 12

2018

Angular Effects on Single-Event Mechanisms in Bulk FinFET Technologies

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Nsengiyumva, Patrick; Massengill, Lloyd W.; Kauppila, Jeffrey S.; Maharrey, Jeffrey A.; Harrington, Rachel C.; Haeffner, Timothy D.; Ball, Dennis R.; Alles, Michael L.; Bhuva, Bharat L.; Holman, W. Timothy; Zhang, En Xia; Rowe, Jason D.; Sternberg, Andrew L.

被引用次数: 26

2018

An Empirical Model for Predicting SE Cross Section for Combinational Logic Circuits in Advanced Technologies

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Jiang, H.; Zhang, H.; Kauppila, J. S.; Massengill, L. W.; Bhuva, B. L.

被引用次数: 5

2018

Time-Domain Modeling of All-Digital PLLs to Single-Event Upset Perturbations

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Chen, Y. P.; Massengill, L. W.; Sternberg, A. L.; Zhang, E. X.; Kauppila, J. S.; Yao, M.; Amort, A. L.; Bhuva, B. L.; Holman, W. T.; Loveless, T. D.

被引用次数: 4

2018

The Impact of Charge Collection Volume and Parasitic Capacitance on SEUs in SOI- and Bulk-FinFET D Flip-Flops

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Ball, D. R.; Alles, M. L.; Kauppila, J. S.; Harrington, R. C.; Maharrey, J. A.; Nsengiyumva, P.; Haeffner, T. D.; Rowe, J. D.; Sternberg, A. L.; Zhang, E. X.; Bhuva, B. L.; Massengill, L. W.

Impact of Single-Event Transient Duration and Electrical Delay at Reduced Supply Voltages on SET Mitigation Techniques

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Maharrey, J. A.; Kauppila, J. S.; Harrington, R. C.; Nsengiyumva, P.; Ball, D. R.; Haeffner, T. D.; Zhang, E. X.; Bhuva, B. L.; Holman, W. T.; Massengill, L. W.

被引用次数: 14

2018

Frequency Dependence of Heavy-Ion-Induced Single-Event Responses of Flip-Flops in a 16-nm Bulk FinFET Technology

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Zhang, H.; Jiang, H.; Bhuva, B. L.; Kauppila, J. S.; Holman, W. T.; Massengill, L. W.

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2018

Exploiting Parallelism and Heterogeneity in a Radiation Effects Test Vehicle for Efficient Single-Event Characterization of Nanoscale Circuits

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Kauppila, J. S.; Maharrey, J. A.; Harrington, R. C.; Haeffner, T. D.; Nsengiyumva, P.; Ball, D. R.; Sternberg, A. L.; Zhang, E. X.; Bhuva, B. L.; Massengill, L. W.

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2018

Effect of Transistor Variants on Single-Event Transients at the 14-/16-nm Bulk FinFET Technology Generation

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Harrington, R. C.; Maharrey, J. A.; Kauppila, J. S.; Nsengiyumva, P.; Ball, D. R.; Haeffner, T. D.; Zhang, E. X.; Bhuva, B. L.; Massengill, L. W.

被引用次数: 17

2018

Effects of Total-Ionizing-Dose Irradiation on Single-Event Response for Flip-Flop Designs at a 14-/16-nm Bulk FinFET Technology Node

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Zhang, H.; Jiang, H.; Fan, X.; Kauppila, J. S.; Chatterjee, I.; Bhuva, B. L.; Massengill, L. W.

被引用次数: 6

2018

Analysis of Bulk FinFET Structural Effects on Single-Event Cross Sections

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Nsengiyumva, Patrick; Massengill, Lloyd W.; Alles, Michael L.; Bhuva, Bharat L.; Ball, Dennis R.; Kauppila, Jeffrey S.; Haeffner, Timothy D.; Holman, W. Timothy; Reed, Robert A.

被引用次数: 45

2017

Effects of Threshold Voltage Variations on Single-Event Upset Response of Sequential Circuits at Advanced Technology Nodes

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Zhang, Hangfang; Jiang, Hui; Assis, Thiago R.; Mahatme, Nihaar N.; Narasimham, Balaji; Massengill, Lloyd W.; Bhuva, Bharat L.; Wen, Shi-Jie; Wong, Richard

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2017

Single-Event Performance of Sense-Amplifier Based Flip-Flop Design in a 16-nm Bulk FinFET CMOS Process

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Jiang, H.; Zhang, H.; Assis, T. R.; Narasimham, B.; Bhuva, B. L.; Holman, W. T.; Massengill, L. W.

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Angular Effects of Heavy-Ion Strikes on Single-Event Upset Response of Flip-Flop Designs in 16-nm Bulk FinFET Technology

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Zhang, Hangfang; Jiang, Hui; Assis, Thiago R.; Ball, Dennis R.; Narasimham, Balaji; Anvar, Ali; Massengill, Lloyd W.; Bhuva, Bharat L.

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2017

Persistent Laser-Induced Leakage in a 20 nm Charge-Pump Phase-Locked Loop (PLL)

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Chen, Y. P.; Loveless, T. D.; Sternberg, A. L.; Zhang, E. X.; Kauppila, J. S.; Bhuva, B. L.; Holman, W. T.; Alles, M. L.; Reed, R. A.; McMorrow, D.; Schrimpf, R. D.; Massengill, L. W.

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2017

Total Dose Measurement Circuit Design Based on a Voltage Reference Topology

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Shetler, K. J.; Holman, W. T.; Kauppila, J. S.; Witulski, A. F.; Bhuva, B. L.; Zhang, E. X.; Massengill, L. W.

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2017

Impact of Temporal Masking of Flip-Flop Upsets on Soft Error Rates of Sequential Circuits

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Chen, R. M.; Mahatme, N. N.; Diggins, Z. J.; Wang, L.; Zhang, E. X.; Chen, Y. P.; Liu, Y. N.; Narasimham, B.; Witulski, A. F.; Bhuva, B. L.; Fleetwood, D. M.

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2017

Charge-Steering Latch Design in 16 nm FinFET Technology for Improved Soft Error Hardness

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Narasimham, Balaji; Chandrasekharan, Karthik; Wang, Jung K.; Ni, Kai; Bhuva, Bharat L.; Schrimpf, Ronald D.

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2017

Estimating Single-Event Logic Cross Sections in Advanced Technologies

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Harrington, R. C.; Kauppila, J. S.; Warren, K. M.; Chen, Y. P.; Maharrey, J. A.; Haeffner, T. D.; Loveless, T. D.; Bhuva, B. L.; Bounasser, M.; Lilja, K.; Massengill, L. W.

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2017

Single-Event Upset Characterization of Common First- and Second-Order All-Digital Phase-Locked Loops

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Chen, Y. P.; Massengill, L. W.; Kauppila, J. S.; Bhuva, B. L.; Holman, W. T.; Loveless, T. D.

被引用次数: 1

2017

Evaluation of SEU Performance of 28-nm FDSOI Flip-Flop Designs

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Wang, H. -B.; Kauppila, J. S.; Lilja, K.; Bounasser, M.; Chen, L.; Newton, M.; Li, Y. -Q.; Liu, R.; Bhuva, B. L.; Wen, S. -J.; Wong, R.; Fung, R.; Baeg, S.; Massengill, L. W.

被引用次数: 29

2017

Effects of Total-Ionizing-Dose Irradiation on SEU- and SET-Induced Soft Errors in Bulk 40-nm Sequential Circuits

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Chen, R. M.; Diggins, Z. J.; Mahatme, N. N.; Wang, L.; Zhang, E. X.; Chen, Y. P.; Liu, Y. N.; Narasimham, B.; Witulski, A. F.; Bhuva, B. L.; Fleetwood, D. M.

被引用次数: 17

2017

Effects of Temperature and Supply Voltage on SEU- and SET-Induced Errors in Bulk 40-nm Sequential Circuits

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Chen, R. M.; Diggins, Z. J.; Mahatme, N. N.; Wang, L.; Zhang, E. X.; Chen, Y. P.; Zhang, H.; Liu, Y. N.; Narasimham, B.; Witulski, A. F.; Bhuva, B. L.; Fleetwood, D. M.

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2017

Single-Event Transient Sensitivity Evaluation of Clock Networks at 28-nm CMOS Technology

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Wang, H. -B.; Mahatme, N.; Chen, L.; Newton, M.; Li, Y. -Q.; Liu, R.; Chen, M.; Bhuva, B. L.; Lilja, K.; Wen, S. -J.; Wong, R.; Fung, R.; Baeg, S.

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2016

An Area Efficient Stacked Latch Design Tolerant to SEU in 28 nm FDSOI Technology

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Wang, H. -B.; Chen, L.; Liu, R.; Li, Y. -Q.; Kauppila, J. S.; Bhuva, B. L.; Lilja, K.; Wen, S. -J.; Wong, R.; Fung, R.; Baeg, S.

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2016

A Comparison of the SEU Response of Planar and FinFET D Flip-Flops at Advanced Technology Nodes

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Nsengiyumva, Patrick; Ball, Dennis R.; Kauppila, Jeffrey S.; Tam, Nelson; McCurdy, Mike; Holman, W. Timothy; Alles, Michael L.; Bhuva, Bharat L.; Massengill, Lloyd W.

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2016

The Contribution of Low-Energy Protons to the Total On-Orbit SEU Rate

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Dodds, N. A.; Martinez, M. J.; Dodd, P. E.; Shaneyfelt, M. R.; Sexton, F. W.; Black, J. D.; Lee, D. S.; Swanson, S. E.; Bhuva, B. L.; Warren, K. M.; Reed, R. A.; Trippe, J.; Sierawski, B. D.; Weller, R. A.; Mahatme, N.; Gaspard, N. J.; Assis, T.; Austin, R.; Weeden-Wright, S. L.; Massengill, L. W.; Swift, G.; Wirthlin, M.; Cannon, M.; Liu, R.;

Chen, L.; Kelly, A. T.; Marshall, P. W.; Trinczek, M.; Blackmore, E. W.; Wen, S. -J.; Wong, R.; Narasimham, B.; Pellish, J. A.; Puchner, H.

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2015

Radiation Hardening of Voltage References Using Chopper Stabilization

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Shetler, K. J.; Atkinson, N. M.; Holman, W. T.; Kauppila, J. S.; Loveless, T. D.; Witulski, A. F.; Bhuva, B. L.; Zhang, E. X.; Massengill, L. W.

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2015

Estimation of Single-Event-Induced Collected Charge for Multiple Transistors Using Analytical Expressions

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Assis, T. R.; Ni, K.; Kauppila, J. S.; Bhuva, B. L.; Schrimpf, R. D.; Massengill, L. W.; Wen, S.; Wong, R.; Slayman, C.

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2015

Single-Event Characterization of Bang-bang All-digital Phase-locked Loops (ADPLLs)

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Chen, Y. P.; Massengill, L. W.; Bhuva, B. L.; Holman, W. T.; Loveless, T. D.; Robinson, W. H.; Gaspard, N. J.; Witulski, A. F.

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2015

Influence of Voltage and Particle LET on Timing Vulnerability Factors of Circuits

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Mahatme, N. N.; Rui, L.; Wang, H.; Chen, L.; Bhuva, B. L.; Robinson, W. H.; Massengill, L. W.; Lilja, K.; Bounasser, M.; Wen, S. -J.; Wong, R.

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2015

Multi-Cell Soft Errors at Advanced Technology Nodes

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Bhuva, B. L.; Tam, N.; Massengill, L. W.; Ball, D.; Chatterjee, I.; McCurdy, M.; Alles, M. L.

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2015

Bias Dependence of Single-Event Upsets in 16 nm FinFET D-Flip-Flops

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Narasimham, Balaji; Hatami, Safar; Anvar, Ali; Harris, David M.; Lin, Alvin; Wang, Jung K.; Chatterjee, Indranil; Ni, Kai; Bhuva, Bharat L.; Schrimpf, Ronald D.; Reed, Robert A.; McCurdy, Mike W.

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2015

An SEU-Tolerant DICE Latch Design With Feedback Transistors

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Wang, H. -B.; Li, Y. -Q.; Chen, L.; Li, L. -X.; Liu, R.; Baeg, S.; Mahatme, N.; Bhuva, B. L.; Wen, S. -J.; Wong, R.; Fung, R.

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2015

Single-Event Upset Characterization Across Temperature and Supply Voltage for a 20-nm Bulk Planar CMOS Technology

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Kauppila, J. S.; Kay, W. H.; Haeffner, T. D.; Rauch, D. L.; Assis, T. R.; Mahatme, N. N.; Gaspard, N. J.; Bhuva, B. L.; Alles, M. L.; Holman, W. T.; Massengill, L. W.

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2015

Single-Event Transient Induced Harmonic Errors in Digitally Controlled Ring Oscillators

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Chen, Y. P.; Loveless, T. D.; Maillard, P.; Gaspard, N. J.; Jagannathan, S.; Sternberg, A. L.; Zhang, E. X.; Witulski, A. F.; Bhuva, B. L.; Holman, T. W.; Massengill, L. W.

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2014

Kernel-Based Circuit Partition Approach to Mitigate Combinational Logic Soft Errors

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Mahatme, N. N.; Gaspard, N. J.; Assis, T.; Chatterjee, I.; Loveless, T. D.; Bhuva, B. L.; Robinson, W. H.; Massengill, L. W.; Wen, S. -J.; Wong, R.

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2014

Impact of Technology Scaling on SRAM Soft Error Rates

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Chatterjee, I.; Narasimham, B.; Mahatme, N. N.; Bhuva, B. L.; Reed, R. A.; Schrimpf, R. D.; Wang, J. K.; Vedula, N.; Bartz, B.; Monzel, C.

被引用次数: 75

2014

Geometry Dependence of Total-Dose Effects in Bulk FinFETs

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Chatterjee, I.; Zhang, E. X.; Bhuva, B. L.; Reed, R. A.; Alles, M. L.; Mahatme, N. N.; Ball, D. R.; Schrimpf, R. D.; Fleetwood, D. M.; Linten, D.; Simoen, E.; Mitard, J.; Claeys, C.

被引用次数: 60

2014

An efficient technique to select logic nodes for single event transient pulse-width reduction

MICROELECTRONICS RELIABILITY

Mahatme, Nihaar N.; Chatterjee, Indranil; Patki, Akash; Limbrick, Daniel B.; Bhuva, Bharat L.; Schrimpf, Ronald D.; Robinson, William

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2013

Single-Event Performance and Layout Optimization of Flip-Flops in a 28-nm Bulk Technology

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Lilja, K.; Bounasser, M.; Wen, S. -J.; Wong, R.; Holst, J.; Gaspard, N.; Jagannathan, S.; Loveless, D.; Bhuva, B.

被引用次数: 59

2013

Reliability-Aware Synthesis of Combinational Logic With Minimal Performance Penalty

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Limbrick, Daniel B.; Mahatme, Nihaar N.; Robinson, William H.; Bhuva, Bharat L.

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2013

Experimental Estimation of the Window of Vulnerability for Logic Circuits

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Mahatme, N. N.; Gaspard, N. J.; Jagannathan, S.; Loveless, T. D.; Chatterjee, I.; Bhuva, B. L.; Massengill, L. W.; Schrimpf, R. D.

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Bias Dependence of Total-Dose Effects in Bulk FinFETs

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Chatterjee, I.; Zhang, E. X.; Bhuva, B. L.; Alles, M. A.; Schrimpf, R. D.; Fleetwood, D. M.; Fang, Y-P; Oates, A.

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2013

Single-Event Transient Effect on a Self-Biased Ring-Oscillator PLL and an LC PLL Fabricated in SOS Technology

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Guo, Shita; Li, Jingxiao; Gui, Ping; Ren, Yi; Chen, Li; Bhuva, Bharat L.

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2013

Sensitivity of High-Frequency RF Circuits to Total Ionizing Dose Degradation

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Jagannathan, S.; Loveless, T. D.; Zhang, E. X.; Fleetwood, D. M.; Schrimpf, R. D.; Haeffner, T. D.; Kauppila, J. S.; Mahatme, N.; Bhuva, B. L.; Alles, M. L.; Holman, W. T.; Witulski, A. F.; Massengill, L. W.

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2013

Effect of Device Variants in 32 nm and 45 nm SOI on SET Pulse Distributions

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Maharrey, J. A.; Quinn, R. C.; Loveless, T. D.; Kauppila, J. S.; Jagannathan, S.; Atkinson, N. M.; Gaspard, N. J.; Zhang, E. X.; Alles, M. L.; Bhuva, B. L.; Holman, W. T.; Massengill, L. W.

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2013

Scalability of Capacitive Hardening for Flip-Flops in Advanced Technology Nodes

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Diggins, Z. J.; Gaspard, N. J.; Mahatme, N. N.; Jagannathan, S.; Loveless, T. D.; Reece, T. R.; Bhuva, B. L.; Witulski, A. F.; Massengill, L. W.; Wen, S. -J.; Wong, R.

被引用次数: 10

2013

Technology Scaling Comparison of Flip-Flop Heavy-Ion Single-Event Upset Cross Sections

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Gaspard, N. J.; Jagannathan, S.; Diggins, Z. J.; King, M. P.; Wen, S-J.; Wong, R.; Loveless, T. D.; Lilja, K.; Bounasser, M.; Reece, T.; Witulski, A. F.; Holman, W. T.; Bhuva, B. L.; Massengill, L. W.

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2013

The Quad-Path Hardening Technique for Switched-Capacitor Circuits

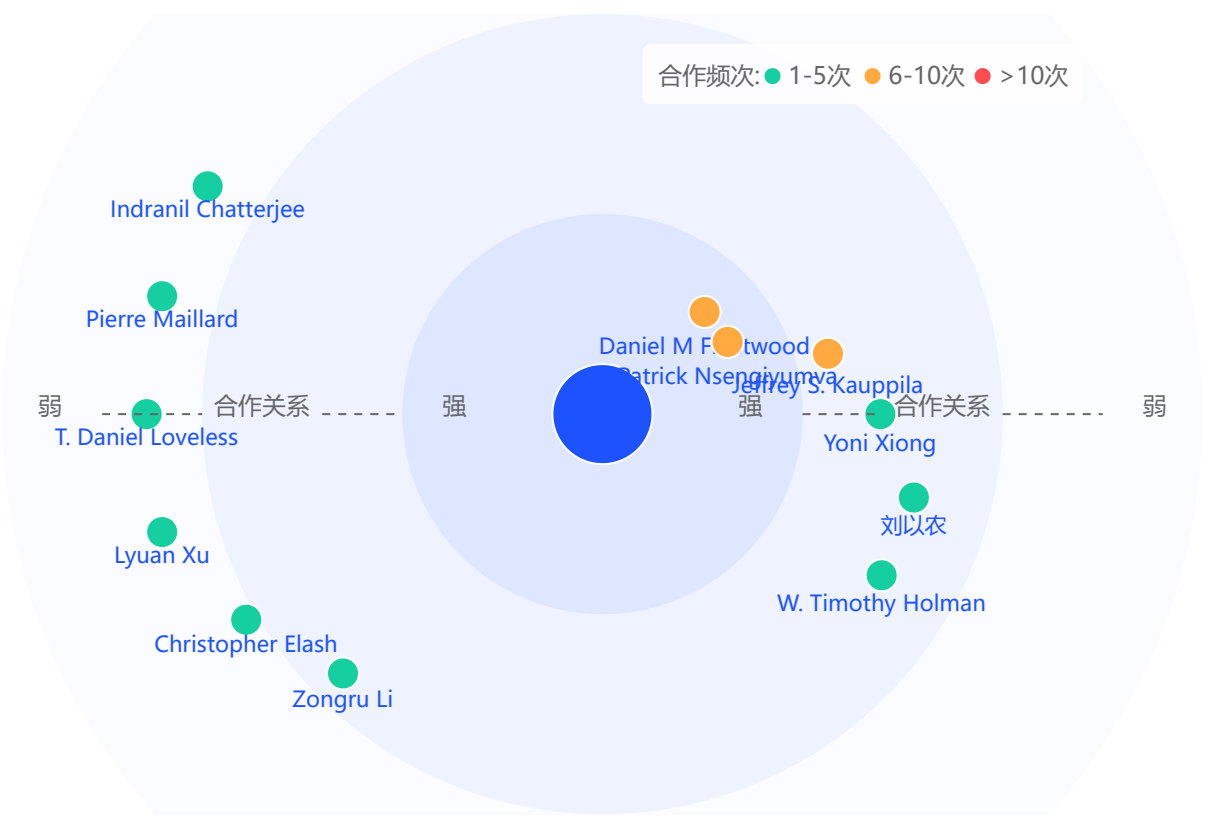
IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Atkinson, Nicholas M.; Holman, W. Timothy; Kauppila, Jeffrey S.; Loveless, T. Daniel; Hooten, Nicholas C.; Witulski, Arthur F.; Bhuva, Bharat L.; Massengill, Lloyd W.; Zhang, En Xia; Warner, Jeffrey H.

被引用次数: 8

2013

学术合作



Daniel M Fleetwood	合作次数：8
范德堡大学	
Patrick Nsengiyumva	合作次数：8
范德堡大学	
Jeffrey S. Kauppila	合作次数：6
范德堡大学	
Yoni Xiong	合作次数：5
范德堡大学	
刘以农	合作次数：4
清华大学	

还有 7 个合作者